Lecture 17

The Central Limit Theorem

(... or "Why is the Gaussian (Normal) distribution so widely used?")

- Recall from previous lectures the following result:

If $X_1, \ldots, X_n$ are independent random variables, and if each X_i, $1 \le i \le n$, is normal with parameters μ and σ^2 ($N(\mu, \sigma^2)$), then the sample mean, $\bar{X}_n = \frac{X_1 + \ldots + X_n}{n}$, has a $N(\mu, \frac{\sigma^2}{n})$ distribution:

$$\bar{X}_n \sim N\left(\mu, \frac{\sigma^2}{n}\right), \text{ or}$$

$$\frac{\bar{X}_n - \mu}{\frac{\sigma}{\sqrt{n}}} = \sqrt{n} \cdot \left(\frac{\bar{X}_n - \mu}{\sigma}\right) \sim N(0,1).$$

Therefore, the c.d.f. of $\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}$ is $\Phi(x)$, the c.d.f. of the $N(0,1)$, that is:

$$\boxed{\mathbb{P}\left(\sqrt{n}\left(\frac{\bar{X}_n - \mu}{\sigma}\right) \le x\right) = \Phi(x), \text{ for Gaussian data}}$$

- We will formalize below the following generalization of the previous result, that will hold for "non-normal data".

- Let $X_1, \ldots, X_n$ be independent, identically distributed (i.i.d.) random variables. Then $\bar{X}_n$ "is approximately" $N(\mu, \frac{\sigma^2}{n})$.

- This result will be valid for ANY distributions with finite means and variances, and will constitute the famous central limit theorem (CLT)

- The precise formulation of "approximately" will involve taking limits as $n \to \infty$. For this reason, the CLT is called "an asymptotic result".

The Central Limit Theorem

Let $X_1, \ldots, X_n$ be i.i.d. random variables with mean μ and variance $\sigma^2 < \infty$. Then, for each number x:

$$\textcircled{1} \quad \lim_{n \to \infty} \mathbb{P}\left(\sqrt{n} \cdot \frac{\bar{X}_n - \mu}{\sigma} \leq x \right) = \Phi(x),$$

where $\Phi(x) = \mathbb{P}(Z \leq x)$, with $Z \sim N(0,1)$, (or, equivalently, where $\Phi(x)$ is the c.d.f of a standard normal variable)

- Interpretation / Jargon: For any distribution, the variable $\bar{X}_n$, when "properly centered" (that is, after we subtract μ) and "properly scaled" (that is, after $\bar{X}_n - \mu$ is multiplied by $\frac{\sqrt{n}}{\sigma}$), will "behave" like a standard Normal random variable, when n is "large".

- Mathematically: The distribution of the r.v

② $\sqrt{n} \cdot \frac{\bar{X}_n - \mu}{\sigma}$ is "very close" to the distribution of Z, when $Z \sim N(0,1)$

- Asymptotic distributions for sums of i.i.d. random variables.

Another way to interpret ① is to re-write ② as follows:

$$\sqrt{n} \cdot \frac{\frac{X_1 + X_2 + \ldots + X_n}{n} - \mu}{\sigma}$$

$$= \sqrt{n} \cdot \frac{(X_1 + \ldots + X_n) - n\mu}{n\sigma}$$

$$= \frac{\sum_{i=1}^{n} X_i - n\mu}{\sqrt{n} \cdot \sigma} \quad \text{and so:}$$

$$P\left(\frac{\sum_{i=1}^{n} X_i - n\mu}{\sqrt{n} \cdot \sigma} \leq x \right) \xrightarrow[n \to \infty]{} \Phi(x) \qquad (3)$$

Then, the CLT also says that:

"The distribution of the sum $\sum_{i=1}^{n} X_i$ is "close" to the distribution of a normal random variable with mean $n\mu$ and variance $(\sqrt{n}\cdot\sigma)^2 = n\cdot\sigma^2$"

when n is large.

Example Let $X_1, \ldots, X_{100}$ be i.i.d. Bernoulli(p) random variables, with $p = \frac{1}{2}$.

Then $X_1 + \ldots + X_{100} \overset{\text{Definition}}{\sim} Y \sim \text{Bin}(n=100, p=\frac{1}{2})$

- Say we want to calculate $P(Y > 51)$.
- "Too much to add up! Use approximations!

Before this, let us see if the CLT can help

We use expression (3) to obtain that:

$$\mathbb{P}\left(\frac{Y - 100\cdot\frac{1}{2}}{10\cdot\sqrt{\frac{1}{4}}} \leq x\right) \approx \Phi(x)$$

which we read as:

Y is <u>approximately</u> normal with mean 50 and standard deviation 5 (or, equivalently, with variance $5^2 = 25$).

- See Figure 6.3: the plot of the probability function of the Binomial$(n=100, p=\frac{1}{2})$ and that of a $N(50, 25)$ are almost indistinguishable.

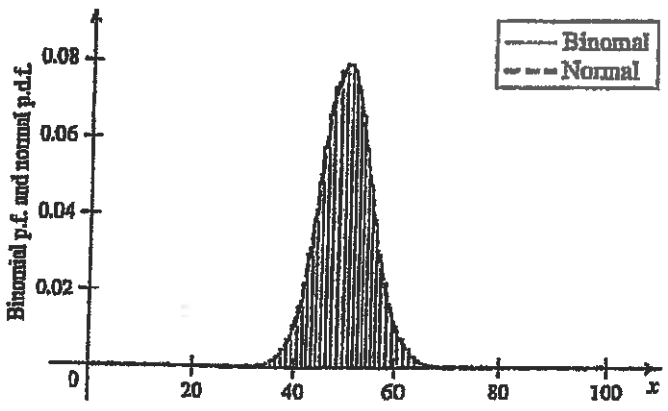


Figure 6.3 Comparison of the binomial p.f. with parameters 100 and 0.5 to the normal p.d.f. with mean 50 and variance 25.

Example

- Let $X_1, \dots, X_n$ iid Uniform $([0,1])$.

  Assume $n = 12$.

  Approximate $\mathbb{P}\left(\left|\overline{X}_n - \frac{1}{2}\right| \leq 0.1\right)$. (***)

  (If this probability is high, this will mean that the average of numbers "drawn" uniformly from $[0,1]$ is about 0.5, plus/minus 0.1).

- We will use the CLT. Recall that it says that:

(*) $\mathbb{P}\left(\dfrac{\overline{X}_n - \mu}{\frac{\sigma}{\sqrt{n}}} \leq x\right) \approx \Phi(x)$, for any $x \in \mathbb{R}$.

From the properties of the normal (random variable) we also know:

(**) $\mathbb{P}\left(\left|\dfrac{\overline{X}_n - \mu}{\frac{\sigma}{\sqrt{n}}}\right| \leq x\right) \approx 2 \cdot \Phi(x) - 1$.

- How to use CLT for our problem:

(i) First calculate μ and σ^2 for $X \sim \text{Uniform}[0,1]$.
This was done in Exercise 3, Section 4.3 (book).

$$\mu = \frac{1}{2} \quad \text{and} \quad \sigma^2 = \frac{1}{12}.$$

(ii) Make sure you know the value of n.
In this case $n = 12$.

(iii) Plug in (**) the values from (i) and (ii) to see what the CLT gives.

$$\mathbb{P}\left(\frac{|\bar{X}_n - \frac{1}{2}|}{\frac{1}{\sqrt{12}} \cdot \frac{1}{\sqrt{12}}} \leq x \right) \approx 2\Phi(x) - 1 \text{ for any } x.$$

$$\Updownarrow$$

(4) $\mathbb{P}\left(12 |\bar{X}_n - \frac{1}{2}| \leq x \right) \approx 2\Phi(x) - 1$ for any x.

- Thus: We can approximate, for any x, any probability like in the LHS of the above line.

• Our problem:

Approximate $\mathbb{P}\left(\left|\overline{X}_n - \frac{1}{2}\right| \le 0.1\right)$

$$= \mathbb{P}\left(12\left|\overline{X}_n - \frac{1}{2}\right| \le 12 \cdot 0.1\right)$$

$$= \mathbb{P}\left(12\left|\overline{X}_n - \frac{1}{2}\right| \le 1.2\right)$$

$$\approx \mathbb{P}\left(|Z| \le 1.2\right) = 2\Phi\left(\tfrac{1}{2}\right) - 1$$

$$\simeq 0.7698.$$

↑ By (4) 2.ed.

• Further jargon related to CLT:

a) The random variable $\sqrt{n}\left(\frac{\overline{X}_n - \mu}{\sigma}\right)$ "converges in distribution" to the standard normal distribution.

b) The **asymptotic** distribution of $\sqrt{n}\left(\frac{\overline{X}_n - \mu}{\sigma}\right)$ is the standard normal distribution.

Approximating the Binomial(n,p) distribution by a normal distribution: The correction for continuity

- Recall that if $X \sim \text{Binomial}(n,p)$, then $\dfrac{X-np}{\sqrt{np(1-p)}}$ is approximately $N(0,1)$.

- We need to "correct" for approximating a discrete distribution by a continuous distribution.

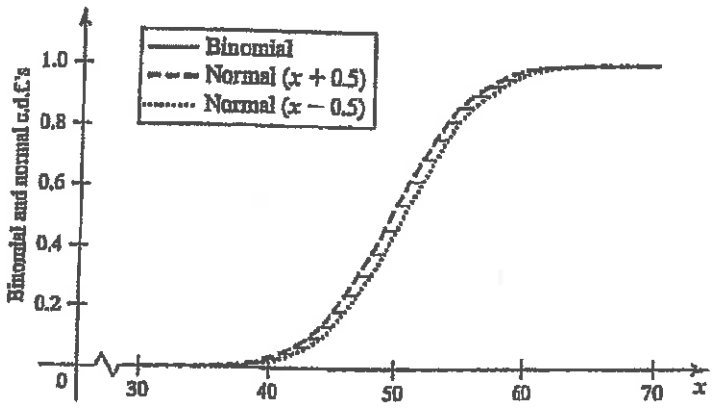
- Let a be a possible value for X.

- Then: $\mathbb{P}(X=a) = \mathbb{P}\left(a-\frac{1}{2} \le X \le a+\frac{1}{2}\right)$

$$\approx \mathbb{P}\left(\underbrace{\frac{a-\frac{1}{2}-np}{\sqrt{np(1-p)}}}_{a_1} \le Z \le \underbrace{\frac{a+\frac{1}{2}-np}{\sqrt{np(1-p)}}}_{a_2}\right)$$

$$= \Phi(a_2) - \Phi(a_1).$$

Figure 6.6 Comparison of binomial c.d.f. with normal c.d.f. shifted to the right and to the left by 0.5.



Example $X \sim \text{Bin}(n=20,\ p=\frac{1}{2})$.

$$\mu = np = 20 \cdot \frac{1}{2} = 10$$

$$\sigma = \sqrt{np(1-p)} = \sqrt{20 \cdot \frac{1}{4}} = 2.236.$$

Then $\mathbb{P}(X=10) = P(9.5 \le X \le 10.5)$

$$= \mathbb{P}\left(\frac{9.5-10}{2.236} \le Z \le \frac{0.5}{2.236}\right)$$

$$\frac{X-10}{2.236} \equiv Z$$

$$\simeq \Phi(0.2236) - \Phi(-0.2236) \simeq 0.1774$$

Additional topics

- Can we make use of the CLT if we are not directly interested in $\bar{X}_n$, but on a function of $\bar{X}_n$?

- Answer: Yes, if the function is appropriately "smooth"

The δ-method (special case)

If h is a function with continuous derivatives, then $\dfrac{\sqrt{n}}{\sigma \cdot h'(\mu)} \cdot [h(\bar{X}_n) - h(\mu)]$ converges in distribution to $N(0,1)$, as $n \to \infty$.

Example

If $X_1, \ldots, X_n$ are i.i.d. rv's from a distribution with mean μ and variance σ^2, what is the approximate distribution of $\frac{1}{\bar{X}_n}$?

Solution Let $h(x) = \frac{1}{x}$. Then $\frac{1}{\bar{X}_n} = h(\bar{X}_n)$. We can use the δ-method: $h'(x) = -\frac{1}{x^2}$, therefore

$$-\frac{\sqrt{n}\cdot\mu^2}{\sigma}\left(\frac{1}{\bar{X}_n} - \frac{1}{\mu}\right) \text{ is approximately } N(0,1)$$

$$\Updownarrow$$

$$\frac{\left(\frac{1}{\mu} - \frac{1}{\bar{X}_n}\right)}{\left(\frac{\sigma}{\sqrt{n}\mu^2}\right)} \text{ is approx } N(0,1)$$

$$\Updownarrow$$

$\frac{1}{\bar{X}_n}$ is approximately normal with mean $= \mu$ and variance $= \frac{\sigma^2}{n\mu^4}$.